NYU TANDON SCHOOL OF ENGINEERING

NYU TANDON CAREER HUB · FALL 2026

Data Science Bootcamp

Week 7 · Probabilistic models



Instructor: **Angelin Ann Jacob**

Section 1 · Tue Nov 10

Section 2 · Nov 12

Useful Links

Zoom
All sessions are virtual. Use ONLY **your NYU SSO Login to be considered for attendance**

Brightspace
Assignments, submissions and the discussion board, monitored Monday to Thursday.

GitHub
Slides and Colab notebooks for every session.

Email
datasciencebootcamp@nyu.edu for questions, extensions and attendance approvals.

1:1 coaching
Optional 30-minute technical interview prep. Book it through Handshake.

THE DATA · DATA STORY 1

Same song, different genre

ONE ROW PER TRACK

32,833

playlist rows

28,352

unique songs

danceability · energy · valence · speechiness · tempo · key · genre

God's Plan
Drake

labelled **pop** **rap**

Smells Like Teen Spirit
Nirvana

labelled **pop** **rock**

460

songs carry more than one genre
Labels made by people are noisy: no model can be 100% right.

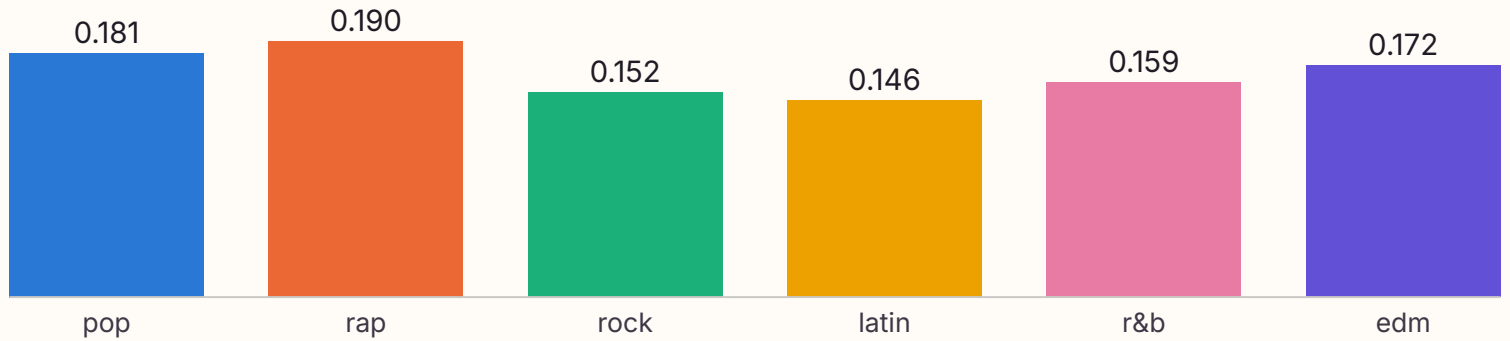
SECTION 1 · EVENTS AND SAMPLE SPACE

Pick a random song: what genre is it?

EXPERIMENT pick a song → **OUTCOME** rock → **SAMPLE SPACE** all 6 genres → **EVENT** rap or r&b

P(GENRE) FOR A RANDOM SONG

adds up to **1.000**



SECTION 1 · PROBABILITY RULES

Count the overlap only once



ADDITION RULE

$$0.190 + 0.158 - 0.058 = 0.290$$

```
P(A or B) = P(A) + P(B) - P(A and B)
```

COMPLEMENT

rap 0.19 | not rap 1 − 0.19 = 0.81

MUTUALLY EXCLUSIVE

rap or rock: no overlap, so just add

Exercise 1 · Probability Basics

SECTION 2 · CONDITIONAL PROBABILITY

Knowing a song is danceable changes the odds

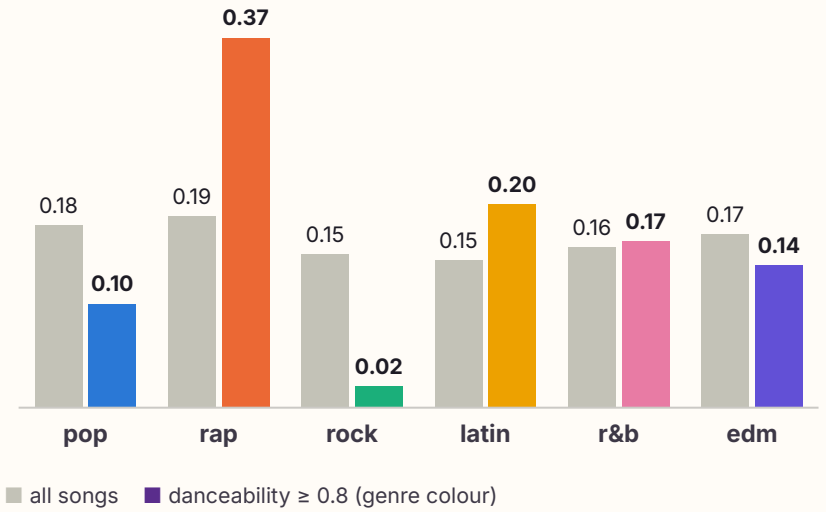
KEEP ONLY THE GIVEN ROW

	rap	not rap
danceable ≥ 0.8	1,646	2,835
less danceable	3,752	20,119

1,646 / 4,481 = 0.367

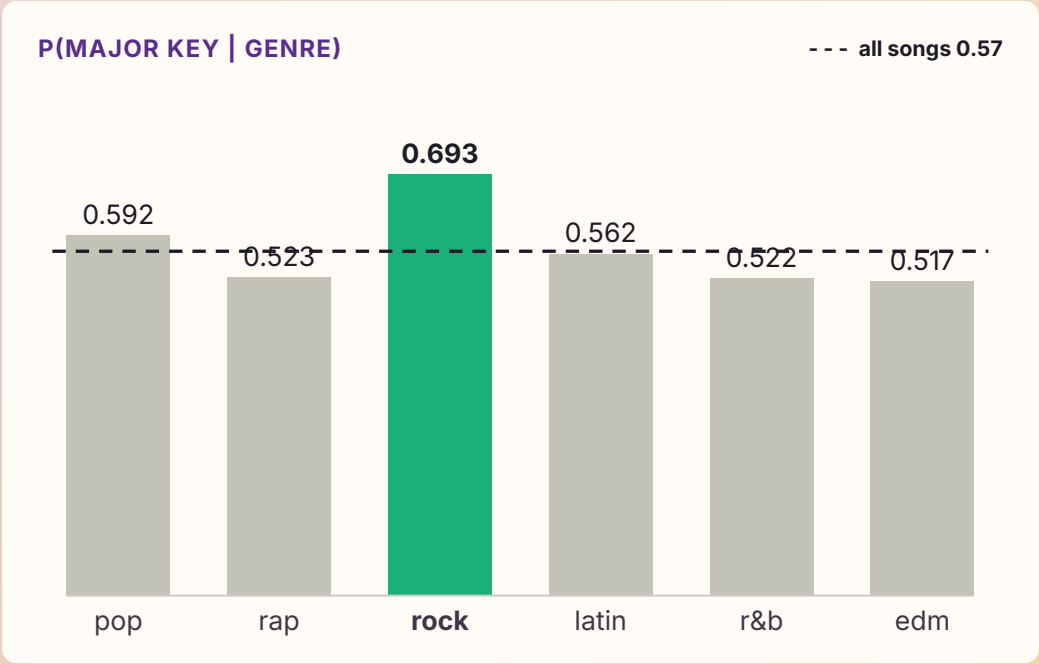
`P(rap | danceable)  vs.  P(rap) = 0.190`

P(GENRE): ALL SONGS VS. DANCEABLE SONGS



SECTION 2 · DATA STORY 2

Rock is the major-key genre



INDEPENDENCE CHECK

`P(A and B) = P(A) × P(B) ?`

rock

0.105 actual

0.086 if independent

× not independent

latin

0.082 actual

0.083 if independent

✓ nearly independent

Exercise 2 · Conditional Probability

SECTION 3 · RANDOM VARIABLES

One playlist, two random variables

SONG 1
Take It All Back 2.0
key C · 148 BPM

SONG 2
Lone Digger
key F · 124 BPM

SONG 3
Dengue Woman Blues
key F · 178.3 BPM

X · SONGS IN THE KEY OF C · DISCRETE

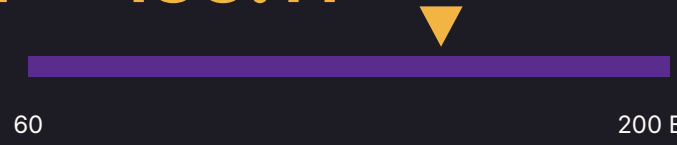
$X = 1$



separate values with gaps

T · AVERAGE TEMPO · CONTINUOUS

$T = 150.11$



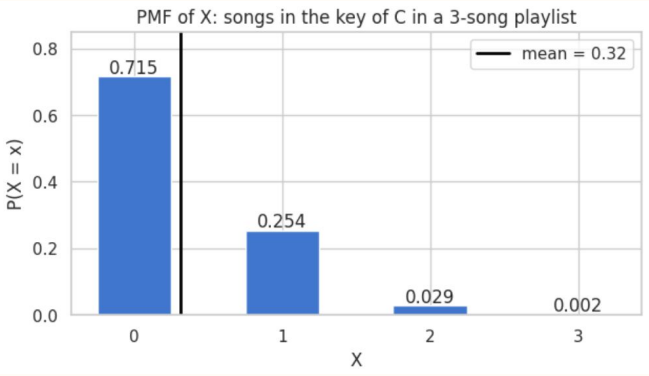
any value in a range

Exercise 3 · Identify Random Variables

SECTION 4 · DISTRIBUTIONS: PMF AND PDF

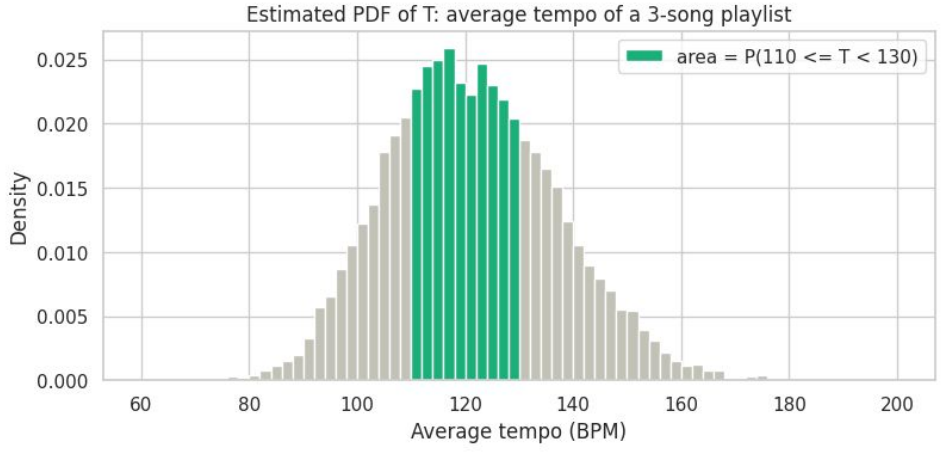
Bars for counts, area for measurements

PMF OF X · HEIGHT = PROBABILITY



bars add up to **1**

PDF OF T · AREA = PROBABILITY

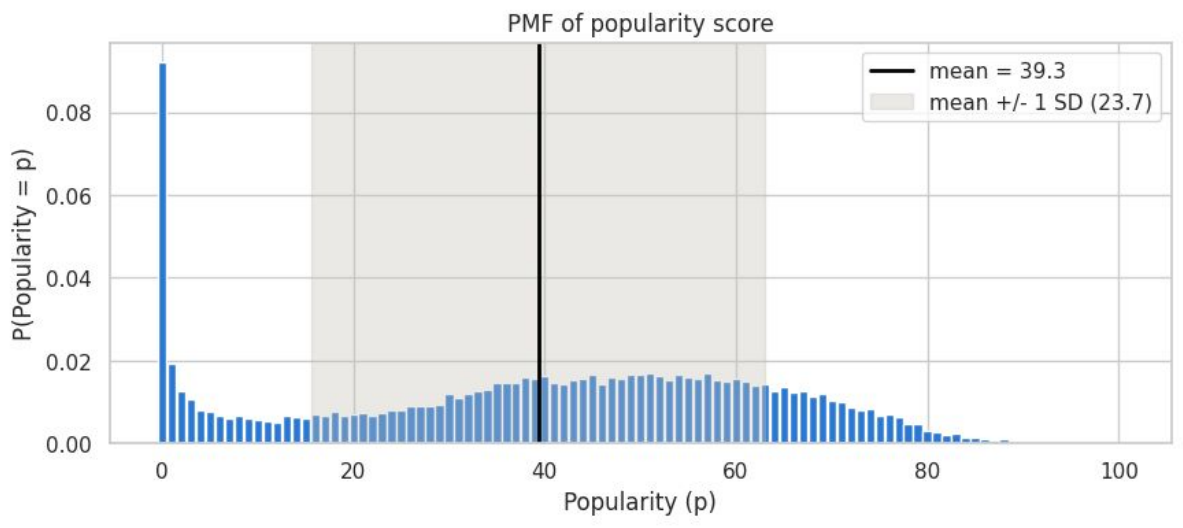


green area = P(110 ≤ T < 130) = **0.468**

SECTION 4 · DATA STORY 3

The songs nobody plays

PMF OF POPULARITY SCORE



9.2%

of songs score exactly 0

39.3

mean popularity

23.7

standard deviation

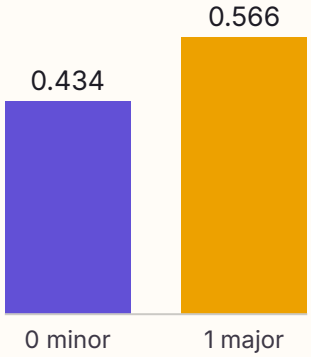
Exercise 4 · Distributions

SECTION 5 · BERNOULLI, BINOMIAL, MULTINOMIAL

Three ways to count

BERNOULLI · 1 SONG

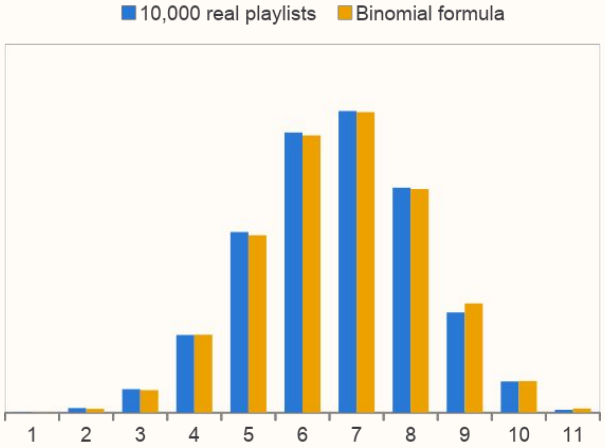
major key?



p = **0.566**

BINOMIAL · 10 SONGS

how many are major?



P(8 or more major) = **0.118**

MULTINOMIAL · 12 SONGS

one random shuffle:



pop 1 rap 4 rock 1
latin 3 r&b 1 edm 2

0.0033

chance of exactly 2 of each genre

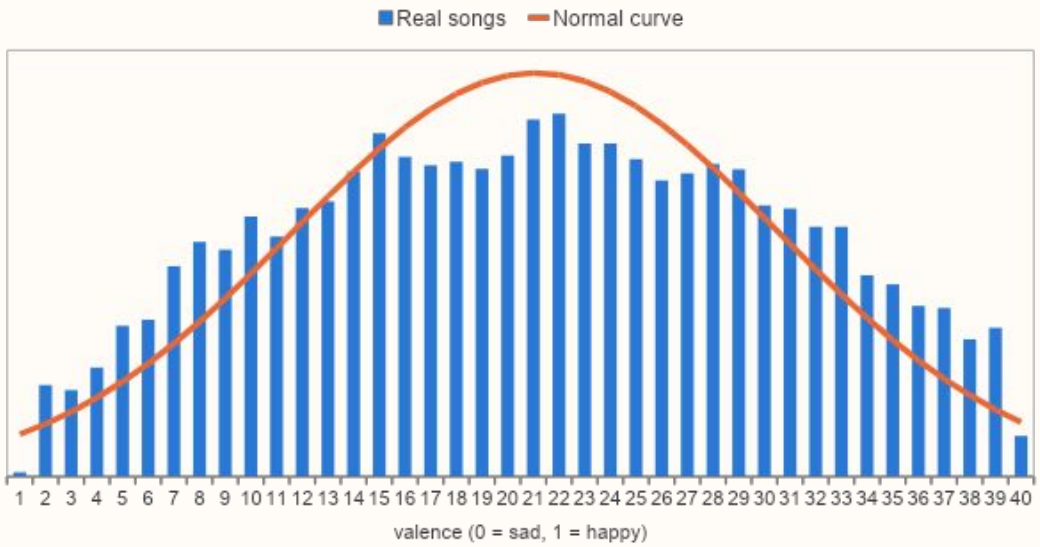
Exercise 5 · Shuffle Math

SECTION 6 · THE NORMAL DISTRIBUTION

Two numbers draw a bell: μ and σ

VALENCE OF 28,352 SONGS

μ = 0.51 σ = 0.23



68-95-99.7 RULE

	rule	real
within 1 SD	68%	62%
within 2 SD	95%	99%
within 3 SD	100%	100%

Z-SCORE · HOW UNUSUAL?

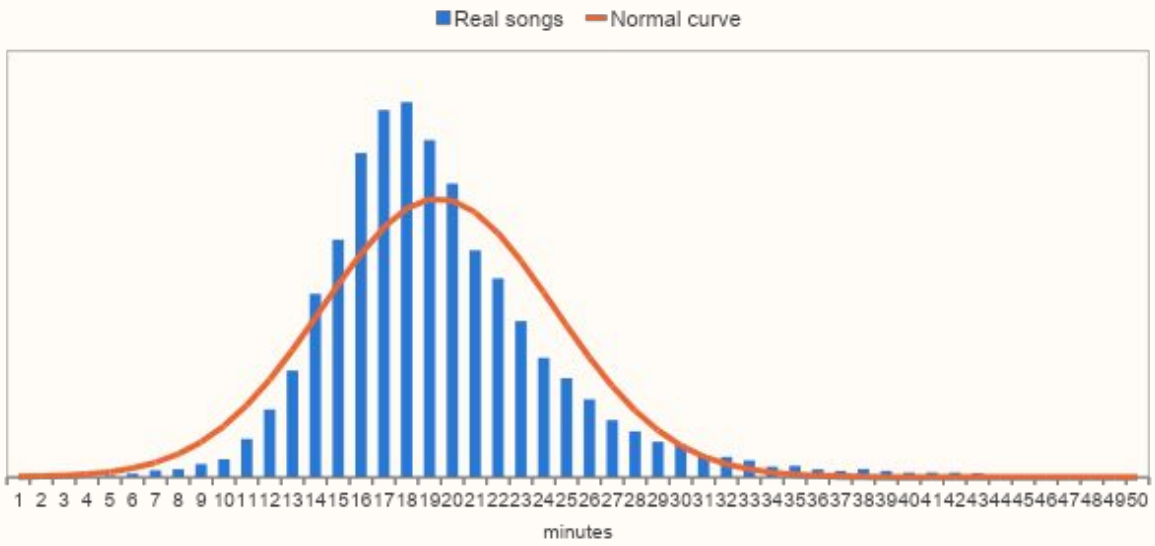
valence 0.9 → z = 1.66

P(> 0.9): model 4.8%, real 4.5%

SECTION 6 · DATA STORY 4

Not every variable is a bell curve

SONG LENGTH IN MINUTES



1.12

skewness (0 = symmetric)

8.6 min

longest song

z = 4.8

a bell says: nearly impossible

Exercise 6 · Your Own Bell Curve

SECTION 7 · BAYES' THEOREM

Update a belief with evidence

POSTERIOR 0.367 — P(rap | danceable)

=

LIKELIHOOD 0.305 — P(danceable | rap)

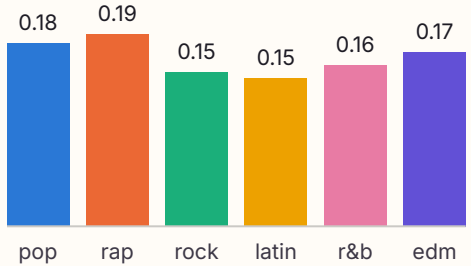
×

PRIOR 0.190 — P(rap)

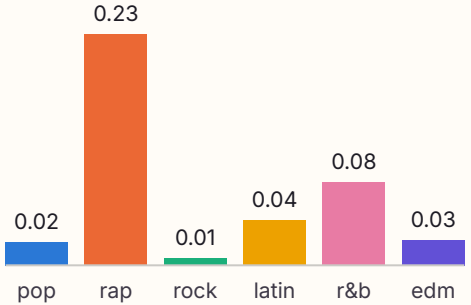
÷

EVIDENCE 0.158 — P(danceable)

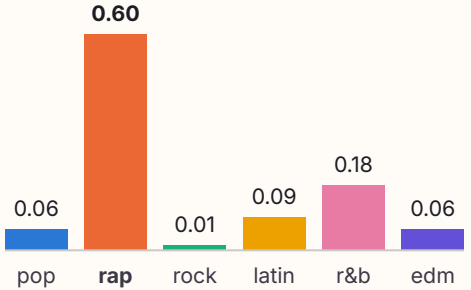
PRIOR · P(GENRE)



LIKELIHOOD · P(SPEECHY | GENRE)



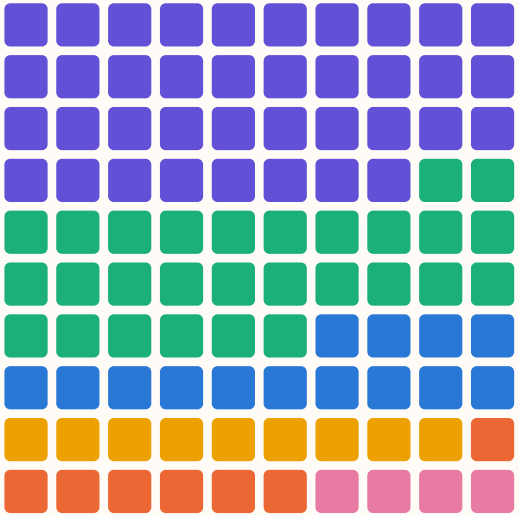
POSTERIOR · P(GENRE | SPEECHY)



SECTION 7 · DATA STORY 5

The base rate trap

100 HIGH-ENERGY SONGS (ENERGY > 0.9)



BEFORE: PRIOR

0.172

P(edm), any song

AFTER: POSTERIOR

0.387

P(edm | energy > 0.9): still under 40%

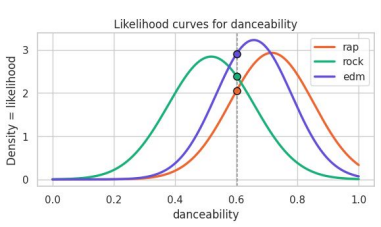
Exercise 7 · Bayes on Spotify

SECTION 8 · NAIVE BAYES

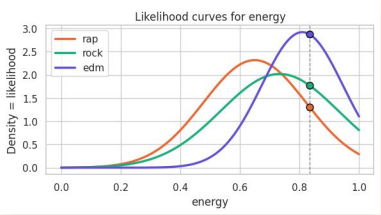
Multiply the evidence, one feature at a time

$$P(\text{genre} \mid \text{song}) \propto P(\text{genre}) \times P(\text{danceability} \mid \text{genre}) \times P(\text{energy} \mid \text{genre})$$

DANCEABILITY · LEVELS = 0.601



ENERGY · LEVELS = 0.834



LEVELS · AVICII · BY HAND

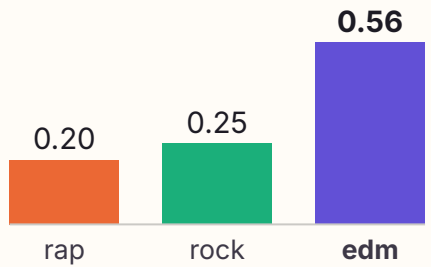
	prior	× dance	× energy	= score	posterior
rap	0.370	2.05	1.31	0.99	0.20
rock	0.295	2.39	1.77	1.25	0.25
edm	0.334	2.90	2.88	2.79	0.56

score ÷ sum of scores = posterior

SECTION 8 · DATA STORY 6

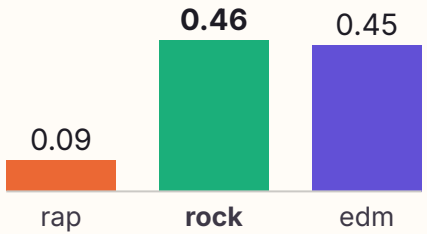
Three famous songs, classified by hand

Levels
Avicii · labelled edm



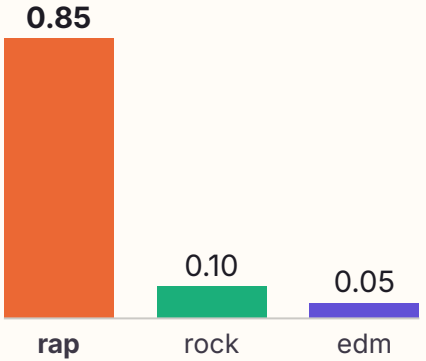
correct

Smells Like Teen Spirit
Nirvana · labelled rock



a close call

God's Plan
Drake · labelled rap



correct

Exercise 8 · Classify a Song

SECTION 9 · NAIVE BAYES VARIANTS

Same recipe, three kinds of input

Test song: **I Would Like - R3hab Remix** labelled edm

GAUSSIAN NB
numbers

energy	0.76
speechiness	0.06
tempo	139.9

POSTERIOR

rap	0.04
rock	0.05
edm	0.92

BERNOULLI NB
yes / no flags

energy > median	1
speechiness > median	0
acousticness > median	1

POSTERIOR

rap	0.11
rock	0.07
edm	0.82

MULTINOMIAL NB
word counts

like	1
r3hab	1
remix	1

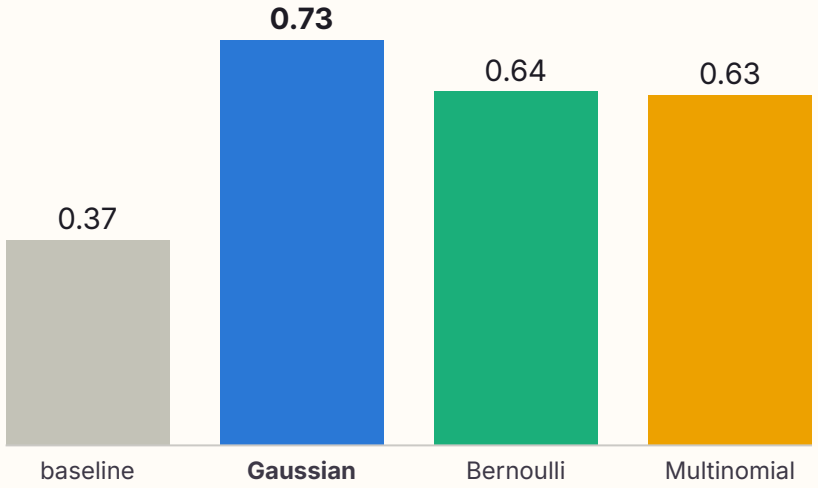
POSTERIOR

rap	0.00
rock	0.00
edm	1.00

SECTION 9 · TEST ACCURACY

All three beat the baseline; Gaussian wins

TEST ACCURACY · RAP, ROCK, EDM



GAUSSIAN NB · CONFUSION MATRIX

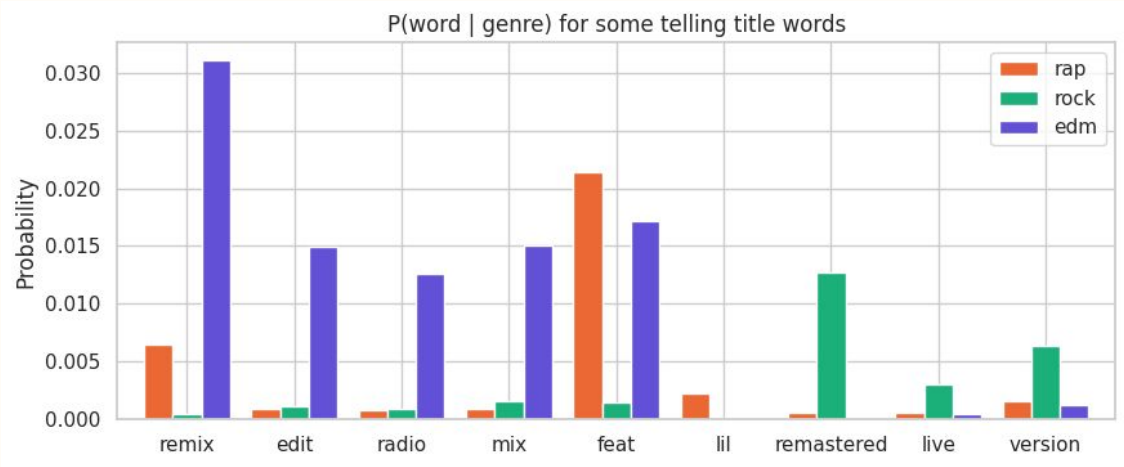
	predicted → rap	rock	edm
rap	0.71	0.11	0.18
rock	0.07	0.68	0.25
edm	0.09	0.10	0.80

rows: actual genre

SECTION 9 · DATA STORY 7

Song titles give the genre away

P(WORD | GENRE) IN TRAINING TITLES



SMOOTHING

'remastered' in training titles

rock	239
rap	9
edm	0

0 × anything = 0

One unseen word would rule out EDM. Laplace smoothing adds 1 to every count.

Exercise 9 · Your Own Classifier

SUMMARY

Eleven ideas from today

Probability

a share of outcomes

Rules

count overlaps once

Conditional

restrict to the given group

Independence

$P(A \mid B) = P(A)$

Random variables

discrete or continuous

PMF vs. PDF

height vs. area

Counting

Bernoulli, Binomial, Multinomial

Normal

μ, σ, z-scores

Bayes

prior × likelihood ÷ evidence

Naive Bayes

multiply per-feature likelihoods

Variants

numbers, flags, word counts

OUR DATA

0.73

Gaussian NB vs. 0.37 baseline

460

songs with two genres